Needle Fear in Community-Based Hemodialysis Patients

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Introduction

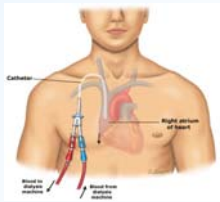
- Needle fears impede crucial efforts to reduce the number of hemodialysis patients who use central venous catheters and to increase the use of permanent vascular accesses, i.e., arteriovenous fistual (AVF) or arteriovenous graft (AVG).
- According to the 2013 United States Renal Data System Annual Report:
  - 80% of incident hemodialysis patients initiated dialysis treatment using a central venous catheter in 2011
  - 63% of incident patients initiated hemodialysis with a catheter alone
  - 52% used a catheter on day 91 of therapy.
- Compared to using a permanent access, central venous catheter use among hemodialysis patients is associated with:
  - greater infection rates (Nassar & Ayus, 2001)
  - a higher risk of hospitalization (Lacson et al., 2010)
  - higher rates of morbidity and mortality (Astor et al., 2005)
  - lower quality of life (Wasse et al., 2007).
- Fear of dialysis needles is one factor associated with high central venous catheter rates among incident hemodialysis patients (Schatell & Merighi, 2010).
- Needle fear is a common condition in the general population (Bienvenu & Eaton, 1998) that inhibits patients from receiving optimal medical care (Öst, 1992).
- In a study of hemodialysis patients, 47% reported that their needle phobia served as a barrier to doing self-care dialysis (McLaughlin et al., 2003).

Study Aim

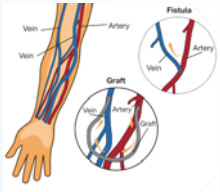
To determine the prevalence of needle fear among in-center hemodialysis patients.

Procedures

- Cross-sectional research design
- Convenience sample drawn from three community-based in-center hemodialysis facilities in New Jersey
- Survey was self-administered or interviewer-administered based on patient preference
- 15 minutes to complete



Central Venous Catheter



Permanent Access (AVF/AVG)

Measure

- 33-item survey developed by the investigators for the Blood, Injection, and Needle Anxiety in Dialysis Treatment (BINA-DT) Project
- 17-item Blood-Injection Symptom Scale (BISS; Page et al., 1997).
- Four dialysis needle fear questions were used for analysis:
  - fear of "seeing" needles
  - fear of "handling" needles
  - fear of "cannulation by other"
  - fear of "self-cannulation."

Participants

- 134 hemodialysis patients
- 63.8% response rate
- 64.9% male
- *N*=97 (72.4% AVF/AVG)
- *N*=37 (27.6% Catheter)
- Dialysis vintage
  - *M*=2.7 years (range = 1 wk to 18 yrs)
- Age
  - *M*=59.6 years (*SD*=14.0)
- Ethnicity
  - 18.8% Hispanic/Latino
- Race
  - 48.4% White
  - 42.1% African American
  - 5.5% Biracial
  - 3.2% Native American
  - 0.8% Asian

Results

- 26% of patients reported fears of "seeing" needles.
- 29% reported fears of "handling" needles.
- More than twice as many catheter patients (30.5%) reported moderate to severe cannulation-by-other fear compared to patients with a permanent vascular access (12.4%), χ^2 (2, N = 120) = 6.97, p = 0.03.
- With regard to self-cannulation, the majority of patients using a catheter (66.7%) or permanent access (62.8%) reported moderate to severe fear.
- Further, a greater proportion of women than men reported moderate to severe self-cannulation fear in the total sample [χ^2 (2, N = 118) = 5.86, p = 0.05] and permanent access subsample [χ^2 (2, N = 94) = 6.52, p = 0.04].



Dialysis Needles

Conclusion

- Dialysis needle fear interferes with patients getting a permanent vascular access, allowing a health care provider to cannulate a permanent access, or electing a dialysis modality such as home hemodialysis that can involve self-cannulation.

Self-Cannulation →



- Findings may inform the development of clinical interventions to reduce dialysis needle fear in an effort to decrease the risks of catheter-related infections and even death.

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